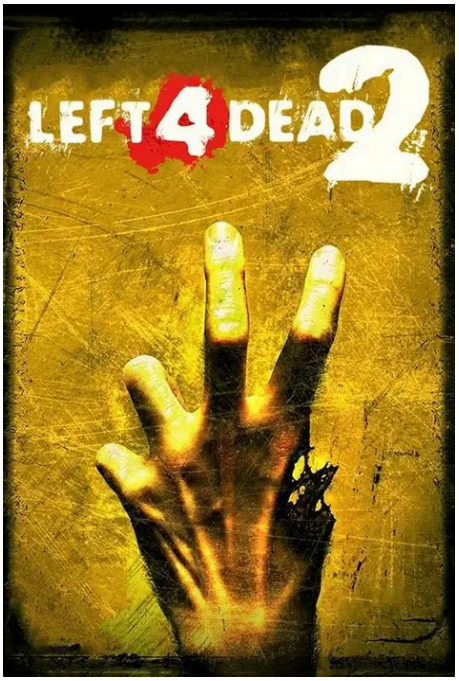
Course: Individual Software Development Process 2026

Iteration Report: Iteration 5

By Left4Dead

Chosen Irl Challenge: Project D: Course Support & Activity Dashboard



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1. Project Snapshot

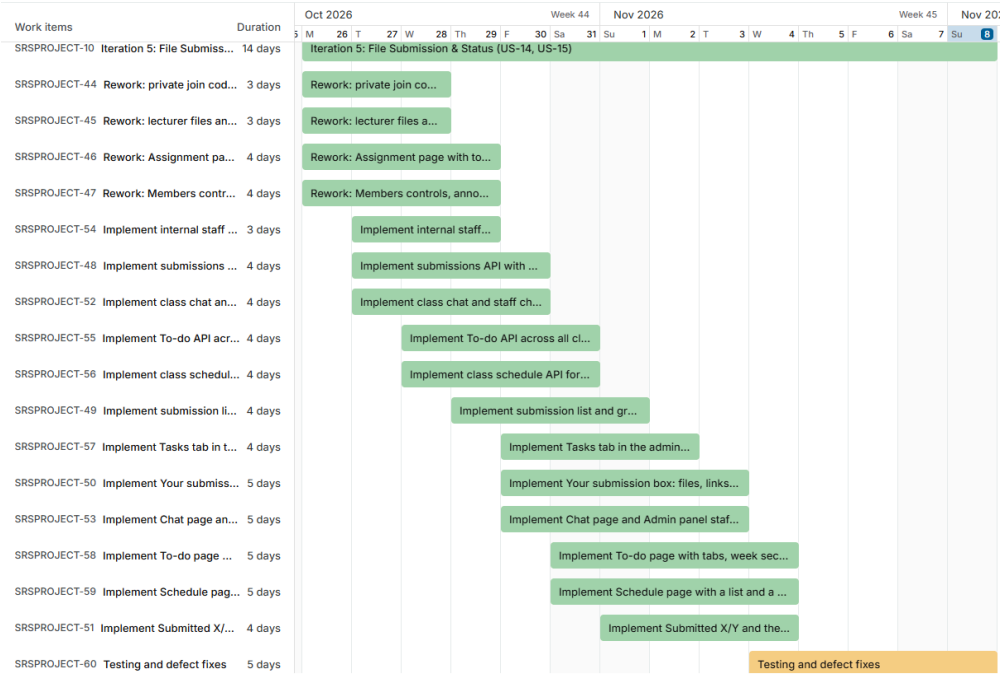
Note: This section must include both Gantt Chart for the whole project and the calculation of the critical path for the project, and each iteration. Created via Plane.so.
Note: Project Repository is mandatory for the report submission of Iteration 2. For Iteration 1, this can leave blank as "(WIP)".
Project Repository: **ThanawinTH/ISP-Left4Dead**

1.1 Project-Wide Gantt Chart



Critical Path: Requirements and SRS Finalization → Architecture and DB Design → Iteration 1: Classroom Setup & Membership (US-1, US-2, US-3) → Iteration 2: Role Assignment and Activity Management (US-4, US-5, US-6) → Iteration 3: Dashboard, Schedule & At-Risk (US-7, US-8, US-9) → Iteration 4: Announcements & Google OAuth2 @ku.th Restriction (US-10, US-11, US-16) → Iteration 5: Staff Channel, File Submission & Grading, Class Schedule, Staff Tasks and To-do (US-12 – US-15, US-17 – US-19) → Iteration 6: Query Search, Admin Role, Automated Testing & Deployment
Critical Path Duration: 98 calendar days

1.2 Current Iteration Gantt Chart



Critical Path: Rework – private join code and member controls → Rework – assignment topics and lecturer attachments → Rework – one Assignment page layout for every role → Implement Submissions API (US-14) → Implement Submission List and Grading (US-15) → Implement Chat API (US-12, US-13) → Implement Staff Tasks API (US-18) → Implement Class Schedule API (US-17) → Implement To-do API (US-19) → Front-end pages for each story → Manual testing & defect fixes
Critical Path Duration: 14 calendar day

2. Sprint/Board Status Snapshot

Note: User Stories Must Be Consistent User Stories in Software Proposal (Week 4). For Maximum Capacity , It Must Be:

- 3 User Stories for Iteration 5

Iteration Goal: Close the communication and submission gaps and bring the system up to the full Project D feature list — students hand work in and staff grade it, the teaching team gets a private channel and a task board, every class activity appears on one schedule, and each member gets a single to-do view across all of their classes. This supports Sub-Goals SG-2, SG-3, SG-4 and SG-5 from the SRS.

Capacity Declared: 4 user stories — US-12 and US-13 carried over from Iteration 4, plus US-14 and US-15 from the project-wide Gantt. Seven were delivered; see Section 3.

User Story	Description	Owner	Status	Within Capacity (Y/N)	Notes
US-12	As a lecturer, TA, or University staff, I want to send a message in the staff channel so that we can settle cover swaps and marking questions without students seeing them.	Pakhin Daonan, Thanawin Thanapornthawan	Done	Y	Carried over from Iteration 4. Backend: chat_messages table with a class channel and a staff channel. Frontend: Chat page and the Admin panel. A student is refused on the staff channel. Covers SRS-18 and SRS-19.
US-13	As a lecturer, TA, or University staff, I want the staff channel to show new messages while I have it open so that a conversation works without reloading the page.	Pakhin Daonan, Thanawin Thanapornthawan	Done	Y	Carried over from Iteration 4. New messages are polled every few seconds and polling pauses while the tab is hidden. Covers SRS-20.
US-14	As a student, I	Pakhin	Done	Y	Backend:

	want to upload my file against an assignment so that my work is recorded before the deadline.	Daonan, Thanawin Thanapornthawan			submissions table, upload API with size and type limits, late marking. Frontend: the Your submission box, files and links submitted together, and Replace Submission. The lecturer can allow late work per assignment. Covers SRS-21 to SRS-24.
US-15	As a lecturer, TA, or University staff, I want to see how many students have submitted so that I can act before the deadline passes.	Phutarak Wongpitak, Kantanut Utamapongchai	Done	Y	Backend: submission list and grades. Frontend: Submitted X/Y on every posted assignment and a grading screen with a file preview. A TA not on the assignment gets the same screen read-only. Covers SRS-25.
US-17	As a lecturer, TA, or University staff, I want to add lab sessions, consultation slots and project milestones alongside assignments and see them	Phutarak Wongpitak, Kantanut Utamapongchai	Done	N	New story, added during the iteration. Backend: timeline_events table. Frontend: the Schedule page with a list view and a month calendar. Assignment due dates

	all on one schedule, so that everything in the course sits in one place.				appear on the same schedule. Not in the submitted SRS — see Section 3.
US-18	As a lecturer, TA, or University staff, I want to create an internal task, assign it to a staff member, change its status and close it, so that work like "prepare demo" is tracked instead of remembered.	Phutarak Wongpitak, Kantanut Utamapongchai	Done	N	New story, added during the iteration. Backend: staff_tasks table, staff-only on read and write. Frontend: the Tasks tab in the Admin panel with Open, In progress and Done columns. Not in the submitted SRS — see Section 3.
US-19	As a lecturer, TA, University staff member or student, I want one page showing what is due and what is at risk across all of my classes, so that I do not have to open each class to find out.	Pakhin Daonan, Thanawin Thanapornthawan	Done	N	New story, added during the iteration. Backend: GET /api/todo across every class the user belongs to, split by their role in each. Frontend: the To-do page with tabs, week sections and a search box. Not in the submitted SRS — see Section 3.

3. Scope Addition/Change Log

Note: Log any item added, expanded, or swapped in after planning began. Empty is a good sign; frequent entries signal scope creep.

Item added/changed	**Reason**	**What as dropped to compensate (one-in-one-out)**
Brought US-12 and US-13 (the staff channel) back in from Iteration 4.ku.th	They were swapped out of Iteration 4 so that US-16 could finally land, and the Iteration 4 report committed them to this iteration. Leaving them a second time would have carried the same story across three iterations.	Nothing was dropped. They were already owed, so they were planned into the declared capacity rather than traded against anything.
Added three user stories that do not exist in the submitted SRS: US-17 (lab sessions, consultations and milestones on one schedule), US-18 (internal staff tasks) and US-19 (what is due and at risk across all my classes).	The Project D requirement list was given to the team after the SRS was submitted. Three of its seven required features — a calendar view combining deadlines, lab sessions, consultation slots and milestones; task and ticket tracking for internal tasks; and an overview dashboard flagging upcoming or at-risk items — had no user story behind them at all. Leaving them to Iteration 6 would have put three required features in the final week.	Nothing was dropped. The iteration ran at seven stories against a declared capacity of four instead, which is the largest process failure of this iteration and is recorded in the retrospective.
Added a rework pass across the front end and parts of the back end before any new story started: a private join code that can be reset,	The Iteration 4 review showed the pages had drifted apart as each story was added to them,	Nothing was dropped. The work was absorbed into the iteration and accounts for four of the sixteen pushes.

member removal, a required subject code, assignment topics, lecturer file and link attachments, and one Assignment page layout shared by lecturer, TA, University staff and student.	and US-14 and US-15 were both going to land on the Assignment page. It was cheaper to make that page consistent once than to add submission and grading to three diverging role views.	
Two of the four improvements carried over from the Iteration 3 and Iteration 4 reports were delivered inside US-19: the unassigned rule from SRS-12 and the staff-side at-risk view. The automated test in the repository was again not delivered.	The at-risk work came free with the To-do API, which had to apply the same staff rules — overdue, due soon, and no TA with a due date inside seven days — across every class. The automated test had no such carrier and was displaced by the three added stories.	Nothing was dropped to compensate. The automated test is now carried for a third time and is the first action item for Iteration 6.

4. Retrospective

What went well	What didn't go well	Action items for next sprint
Doing the rework pass before any new story paid for itself. Topics, lecturer attachments and one Assignment page layout meant US-14 and US-15 landed on a page that was already consistent, instead of three role views drifting further apart.	The iteration ran at seven user stories against a declared capacity of four. That is not a sustainable pace, and it happened because the gap between our SRS and the Project D requirement list was found late rather than at planning.	Write the SRS addendum for US-17, US-18 and US-19, with their SRS numbers and traceability rows, so that the specification and the implementation describe the same system.
Splitting every story back end first and front end second, as sixteen small pushes each with its own README, meant nobody was blocked for long and any single push could be reverted on its own without touching the rest.	The automated test in the repository was an action item in the Iteration 3, Iteration 4 and Iteration 5 reports and still does not exist. Every story in this iteration was tested by hand, at the end.	Write the three refusal tests — a non-@ku.th sign-in, a student getting 403 on create, and a blank title getting 400 — in the first two days of Iteration 6, before any new feature is started.
Three required features that had no user story at all were found and closed inside one iteration, so the system now covers five of the seven features in the Project D requirement list rather than two.	Two required features are still open: activities and tasks cannot be query-searched from the server, and the membership role enum still has no admin value, so the Admin role named in the requirement does not exist.	Close both remaining requirement bullets early in Iteration 6: one GET /api/search endpoint across activities, tasks and announcements, and an admin value added to the membership role enum with the Members page updated to use it.

5. Testing & Quality Notes

D-01 — Database schema synchronization: The Announcements page displayed "Server error" on a developer machine because an existing MySQL volume did not contain the newly added announcements table. Fixed by recreating the volume. The team updated the push instructions to require `docker compose down -v` followed by `docker compose up --build` after pushes that change the database schema, including submissions, grades, chat_messages, staff_tasks, and timeline_events.

B-02 — Assignment title validation: The automated boundary test found that an assignment title exceeding 190 characters returned HTTP 500 instead of the expected HTTP 400. **Status: WIP** — The team is currently working on fixing this bug and will rerun the test after the fix.

Testing results: All 56 acceptance tests passed, including all 14 Iteration 5 tests. All 7 upload boundary tests passed. Overall, 85 of 86 automated tests passed, with B-02 remaining under investigation.

6. Project Risk Register

Risk ID	Description	Likelihood (L/M/H)	Impact (L/M/H)	Mitigation/ Contingency	Status
R-01	Some tasks depend on other tasks and may cause delays.	M	M	Order the work back end first and front end second, and publish the push order in advance so everyone knows what they are waiting for.	Mitigated
R-02	Frontend and backend may not work together correctly.	L	H	Each back-end push ships before its front-end pair and carries a README stating the endpoints and the response shape, so the front end is written against an agreed contract.	Mitigated
R-03	US-16 (Google login with the @ku.th restriction) was delayed for several iterations and may still have problems.	L	L	Delivered and demonstrated in Iteration 4; retested in this iteration as part of the chat and to-do work, which both depend on the signed-in identity.	Closed
R-04	The submitted SRS no longer describes the system. US-17, US-18 and US-19 are implemented but have no user story, no URS and no SRS requirement behind them.	H	M	Write an SRS addendum in Iteration 6 covering the three stories with their SRS numbers and traceability rows, rather than reissuing the whole document.	Open

7. Individual Contribution Log

Note: One row per member, filled in by that member. The submission of this file considered that the form below is consented and confirmed by the whole team. This section directly supports the Individual Project Contribution and Peer Evaluation grading components.
Note 2: For Iteration 1, evidence can be a reference to the video presentation/iteration report that is related to the responsible task(s) (e.g. Lalita's Evidence). However, for Iteration 2, the reference must be on **GitHub** file and commit only, which is uploaded by that team member.

Student Name	**Tasks owned this sprint**	**Course concept(s) applied**	**Evidence**	**Self-rated effort (1-5)**
Thanawin Thanapornthawan	Rework of the Assignment page; US-14 front end; US-12 and US-13 front end; US-19 front end	Role-appropriate views from one layout; validate in the browser and again on the server; polling for updates without a reload	js/schedule.js, js/api.js — e124618; js/schedule.js, js/create-assignment.js — 282b648; chat.html, admin.html, js/chat.js — 94e2c52; todo.html, js/todo.js — [commit hash, push 16]	5
Phutarak Wongpitak	Rework of join code and member controls; US-15 back end; US-18 back end; US-17 back end	Least privilege on every route; server-set author and timestamp; referential integrity when a member is removed or demoted	src/classrooms.js, src/members.js — a9e5e2e; src/submissions.js, db/schema.sql — dafbe0a; src/tasks.js — [commit hash, push 11]; src/timeline.js — [commit hash, push 13]	5
Pakhin Daonan	Rework of attachments and topics; US-14 back end; US-12 and US-13 back end; US-19 back end	One resource with two rules; size and type validation on the server; a shared risk rule so two pages cannot disagree	src/attachments.js, src/files.js, src/topics.js — 923bdb9; src/submissions.js — 99ea14e; src/chat.js — 95654e3;	5

			src/todo.js — [commit hash, push 15]	
Kantanut Utamapongchai	Rework of Members and Announcements ; US-15 front end; US-18 front end; US-17 front end	Read-only views for roles that may look but not change; clear state on every row; two views over one data set	announcements.html, js/members.js — e172472, 9959785; js/schedule.js — 7e03396; js/admin.js, create-task.html — [commit hash, push 12]; timeline.html, js/timeline.js — [commit hash, push 14]	5